

8086 SYSTEM BUS STRUCTURE:

8086 system signals:

1. 40 pin IC package.

2. 16 bit microprocessor.

3. 20 bit address bus lines.

8086 SIGNALS:

		MAX	MIN
GND	1	40 Vcc	
AD14	2	39 AD15	
AD13	3	38 A16/S3	
AD12	4	37 A17/S4	
AD11	5	36 A18/S5	
AD10	6	35 A19/S6	
AD9	7	34 BHE/S1	
AD8	8	33 MN/MAX	
AD7	9	32 RD	
AD6	10	31 RD/STRO	HOLD
AD5	11	30 RD/STRO	HLDA
AD4	12	29 LOCK	WR
AD3	13	28 S	M/IO
AD2	14	27 S	DT/R
AD1	15	26 S	DEN
AD0	16	25 S	ALE
NMI	17	24 S	INTA
INTERR	18	23 $\overline{INT}$	
CLK	19	22 Ready	
GND	20	21 Reset	

Categories:

[∵ MIN mode is single processor mode]

i). Common signals to both min and max mode configuration.

* If $\overline{MN}/\overline{MX}$ is (high) then it is min mode,
if $\overline{MN}/\overline{MX}$ is set to 0, it is max mode.
(low)

[MAX mode - multi processor mode]

ii). signals specific to minimum mode configuration.

iii). signals specific to maximum mode configuration.

$\Rightarrow$ 8086 work with 5V power supply
is connected to 40th (V_{CC}) pin.

$\Rightarrow$ Ground is connected to 1st (GND) pin.

i). Common signals.

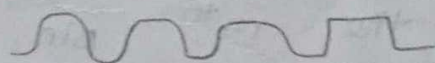
$\Rightarrow$ ADO to AD15 $\rightarrow$ multiplexed address / data lines

[It is 16-bit μ processor]

* In order to reduce the resource,
resource utilization is improved - reason
for multiplexed.

• Machine cycle - time taken to complete
a single instructions.

* It has 4 'T' states.



machine cycle - combination of 4 'T' states

ADO-AD15 :

* During 'T₁' it act as address line,

during other 'T' states it act as data line.

If the cycle cannot complete in the given T states, so we consider a wave b/w T_3 & T_4 as T_6 .

⇒ NMI - Non-maskable Interrupt.

- * It is hardware interrupt pins.

- * It is higher priority, request given by this can't be masked (disable)

- * This NMI is edge triggered input, low to high transition indicates the interrupt response at the end of current instruction.

⇒ INTR - Interrupt pin

- * This is level triggered input.

This is sampled during last clock cycle [T_4] of each instruction to determine the availability of request.

- * It can be disable.

⇒ CLK signals - speed of the processor.

- * To generate wave signal, we use oscillator, to the 19th pin.
[Provides basic timing for processor (bus controller)]

⇒ RESET pin.

- * This input causes the processor [signal is active HIGH, for atleast 4 clock cycles]

to terminate the current activity
and start execution from FFFF0H
⇒ Ready signal. (address)

* This is acknowledgment from
slow devices (or) memory, that they have
completed the data transfer. (signal is active
HIGH)

⇒ Test pin.

* If it goes low execution will
continue else processor remains in an
idle state.

⇒ $\overline{RD}$ - read signal. [low - read
high - write]

* It is the active low signal
during T_2 , T_3 and T_w of any read cycle.
⇒ $MN/\overline{MX}$ pin.

* If it is set to 1 - MIN mode.

* If it is 0 - called as MAX mode.

⇒ A_{19}/S_6 , A_{18}/S_5 , A_{17}/S_4 , A_{16}/S_3 pins.

* It is multiplexed address & status line.

* During T_1 state, it will act as
address line.

* During other time states, it will
act as status line.

S_6 → always low

S_5 → Interrupt enable flag bit.

↘
This will be updated at the
beginning of each clock cycle (T_1)

* S_3 indicates which segment ^{register} is currently used [S_3 & S_4].

S_4	S_3	
0	0	→ All data (All data)
0	1	→ stack
1	0	→ code
1	1	→ data.

⇒ $\overline{BHE} / S_7$ - Bus high enable.

* This is used to indicate transfer of data over higher order data bus.

[data bus size is 16 bit]

↓
[In this 8 bit is lower order ^{data} & 8 bit is higher order]

* active low signal. (during T_1)

S_7 - status line [No purpose specifically]
↳ status information available during T_2, T_3, T_4

ii) MAX mode signals.

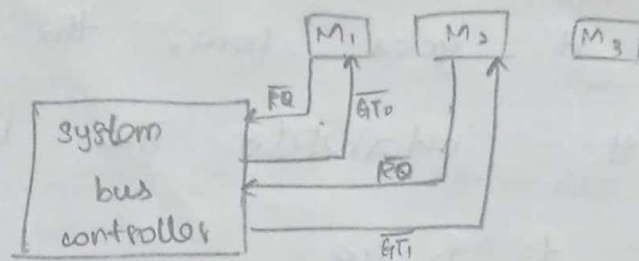
It has more than 1 masters.

[multiple masters]

⇒ $\overline{RQ} / \overline{GT_0}$, $\overline{RQ} / \overline{GT_1}$ - Request Grant signal.

* If any master wanted to access the bus, it sends a Request signal.

* The bus controller sends the granted signal. Then only master get the bus control.



* each pin is bidirectional.

* bus control is in M_1 & M_2 (masters)

$\Rightarrow$ $\overline{Lock}$ signal.

* This signal will prevent other system bus masters from gaining the system bus control.

$\Rightarrow Q_{s1}, Q_{s0}$ - status of code prefetch queue.

$\Rightarrow \overline{S}_2, \overline{S}_1, \overline{S}_0 \Rightarrow$ status lines

0 0 0 $\rightarrow$ INTA [interrupt acknowledgment signal]

0 0 1 $\rightarrow$ Read I/O port

0 1 0 $\rightarrow$ write I/O port.

0 1 1 $\rightarrow$ Halt

1 0 0 $\rightarrow$ code access

1 0 1 $\rightarrow$ read memory [opp to write]

1 1 0 $\rightarrow$ write memory. processor $\xrightarrow[\text{memory to}]{\text{data}}$ memory

1 1 1 $\rightarrow$ passive state.

[active during T_1, T_2 and return to passive state (1,1,1)]
iii) MIN mode signals. T_3 (or) T_w .

$\Rightarrow M/\overline{IO}$ - memory / input output operation.

If $M/\overline{IO} = 1 \Rightarrow$ performs memory operation.

$M/\overline{IO} = 0 \Rightarrow$ perform IO.

$\Rightarrow$ INTA - Interrupt acknowledgement.

* when it goes low, the processor accept the interrupt. This will be low at t_3, t_2, t_w .

⇒ ALE - Address Latch enable.

* It indicates the availability of valid address on data bus.

* Active during ' T_1 ' of any bus cycle.

⇒ $DT/\bar{R}$ - Data transmit / Receive

* If $DT/\bar{R} = 1$, data is transmit

[from processor to memory
[29]]

* If $DT/\bar{R} = 0$, data is received.

[from memory (or) IO to
processor]

⇒ DEN. - Data enable.

* Active low signal.

* This indicates the availability of valid data on address (or) data bus.

* This will be active low from middle of ' T_3 ' to middle of

T_4

→ HOLD, HLDA pin.

* HOLD - hold.

HLDA - hold acknowledgement.

* HOLD = 1 indicates to the processor that another master is requesting the bus access.

* The processor after receiving whole request will issue acknowledgement through HLDA pin.

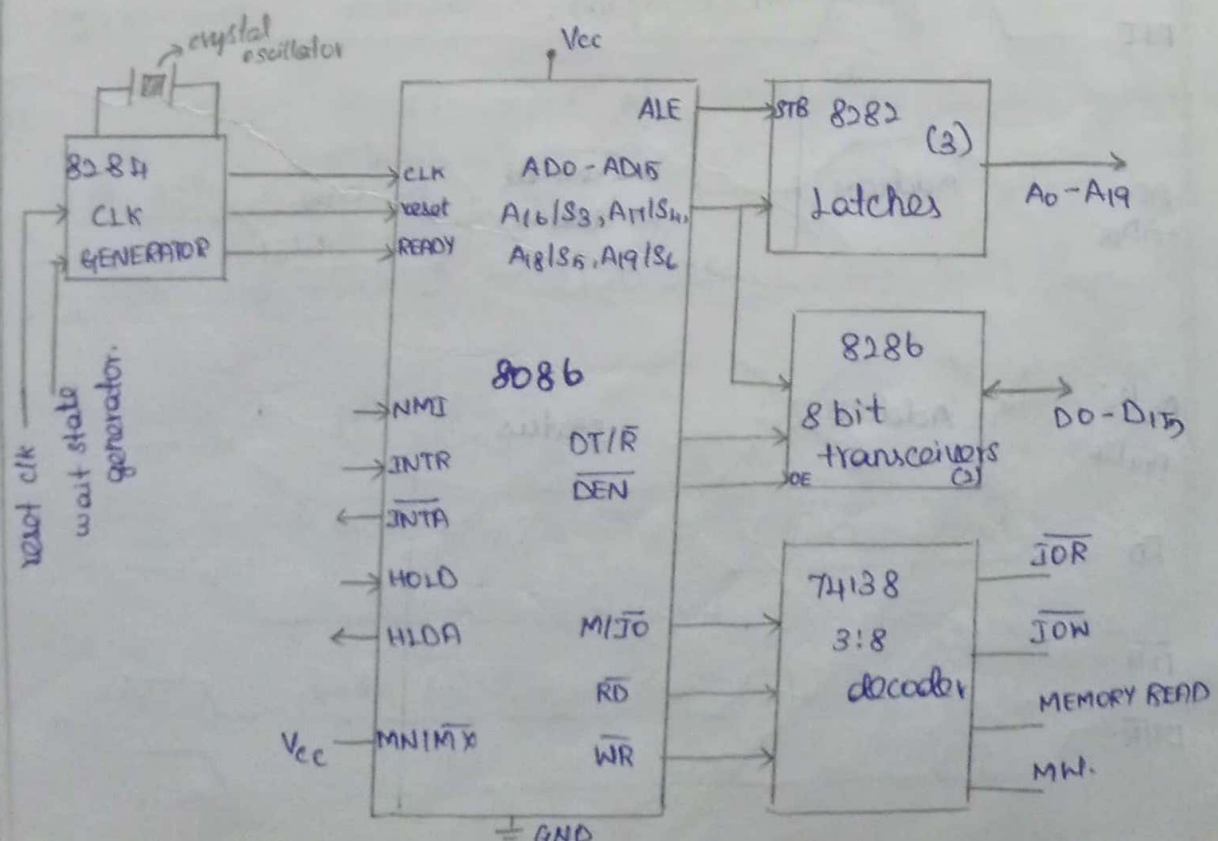
SYSTEM DESIGN USING 8086.

Minimum mode configuration of 8086:

* $MN/\overline{MX} = 1$

* all the control signals generated by 8086 processor.

8282 latch - 8 bit latch (it has D flip/flop)



latch was: In order to separate the address from multiplexed data.

* 8086 processor.

→ 8283 Latches

• ALE signal go high during T_1 , $ALE=1$.

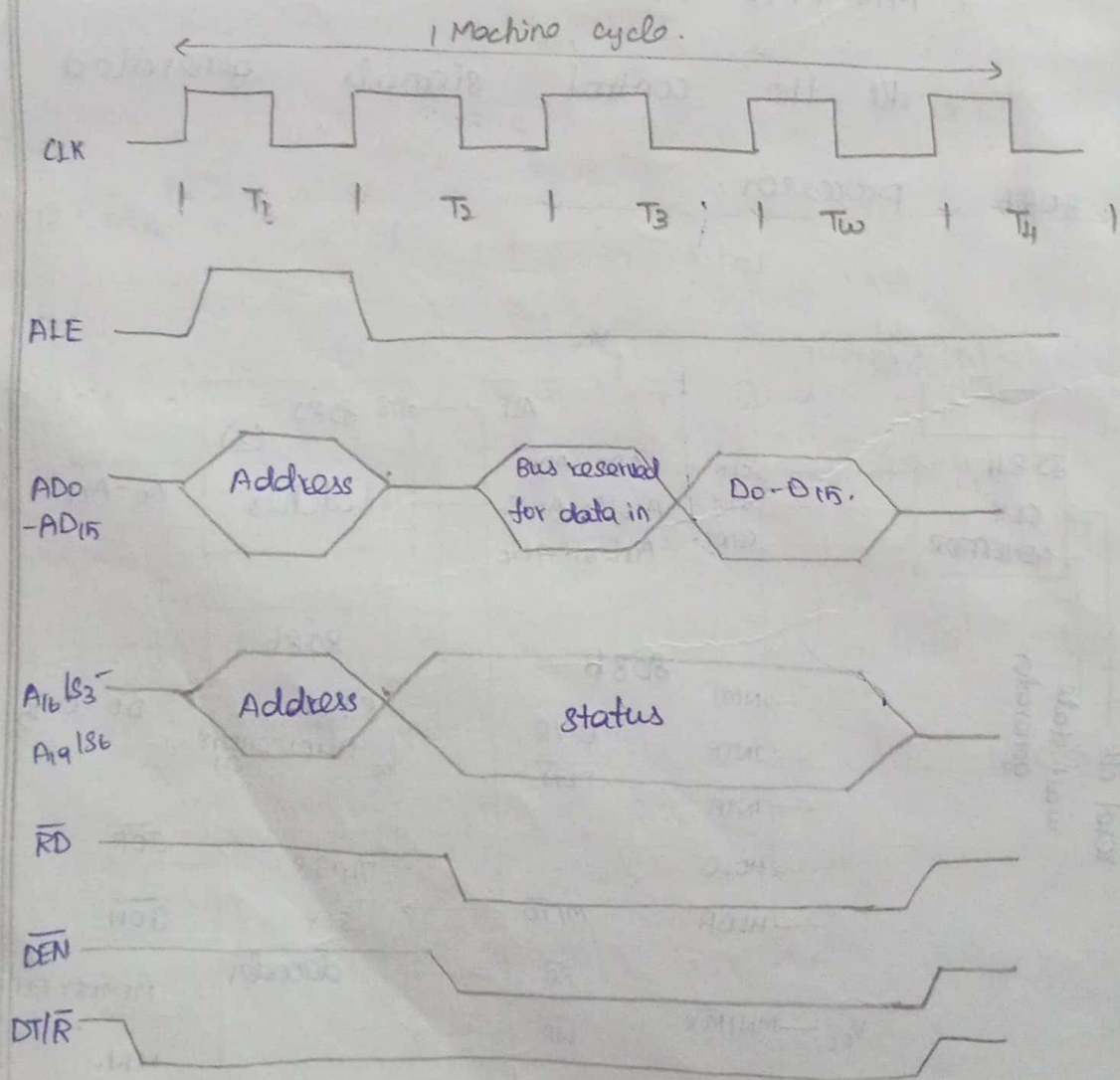
→ 8286 - 8 bit transceiver.

• DEN is connected to data enable.

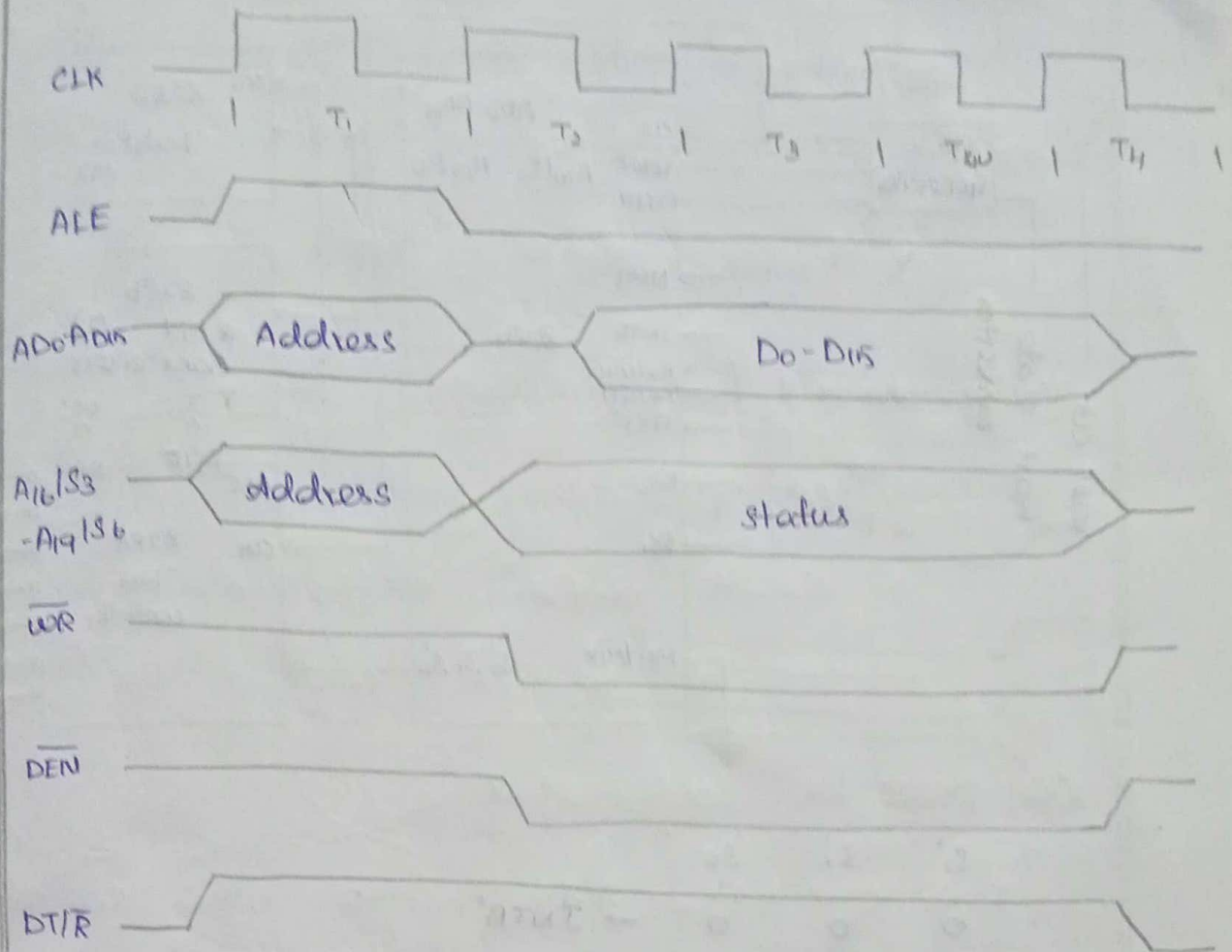
• It is bidirectional.

→ 74138 3:8 decoder.

Memory Read: [Timing diagram of memory Read cycle]



Memory Write:



From T_2 to T_4 $\rightarrow$ data available.

At T_1 $\rightarrow$ address available.

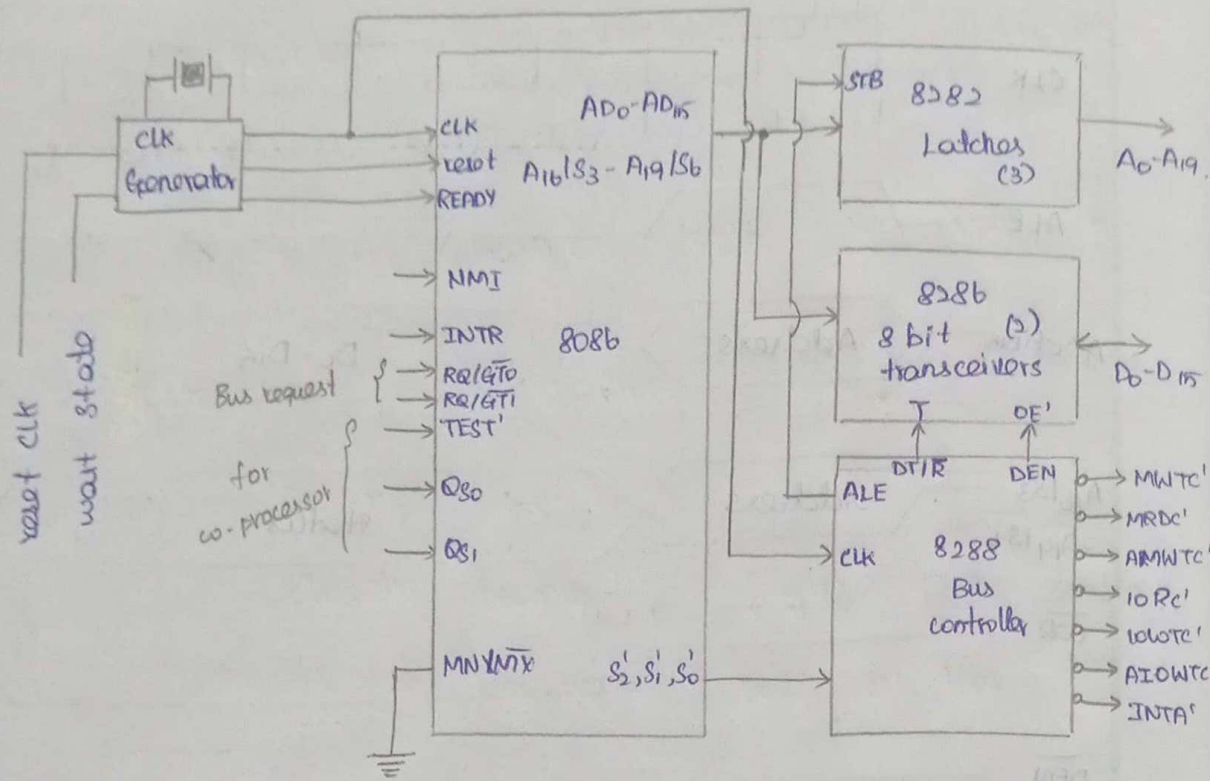
Maximum mode configuration of 8086:

STB $\rightarrow$ strobe input of latch, ALE is connected to STB.

Additional processor is used for

floating pointer.

$\overline{S_2} - \overline{S_0}$ $\rightarrow$ active T_4 to mid point of T_5 .



S_2'	S_1'	S_0	
0	0	0	$\rightarrow$ INTA'
0	0	1	$\rightarrow$ IORC'
0	1	0	$\rightarrow$ IOWC & AIOWC' $\downarrow$ Advance IO write
0	1	1	$\rightarrow$ None.
1	0	0	$\rightarrow$ MWTC
1	0	1	$\rightarrow$ MRDC
1	1	0	$\rightarrow$ MWTC' & AMWTC'
1	1	1	$\rightarrow$ None.

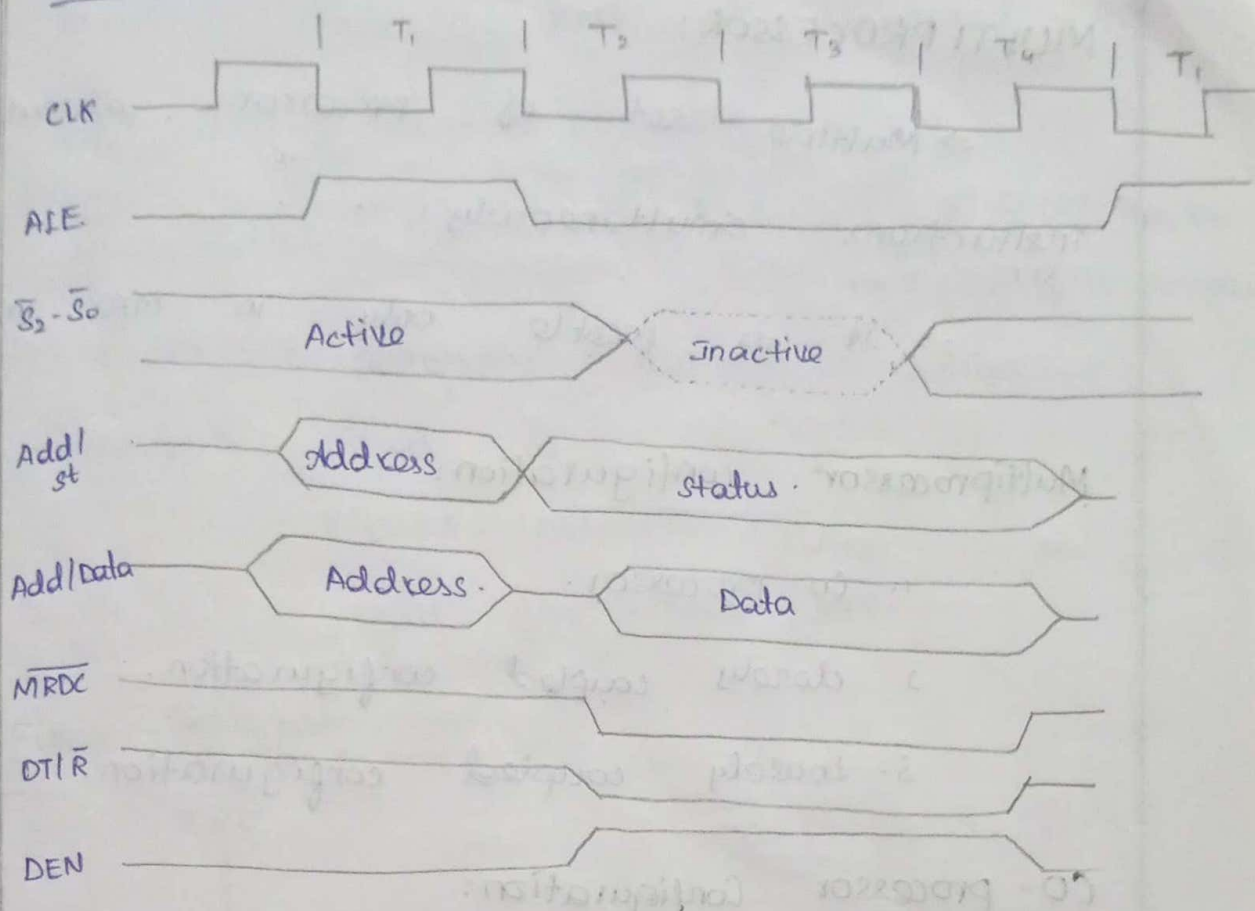
Advantages:

1. Efficiency will improve.
2. It supports multi processing.

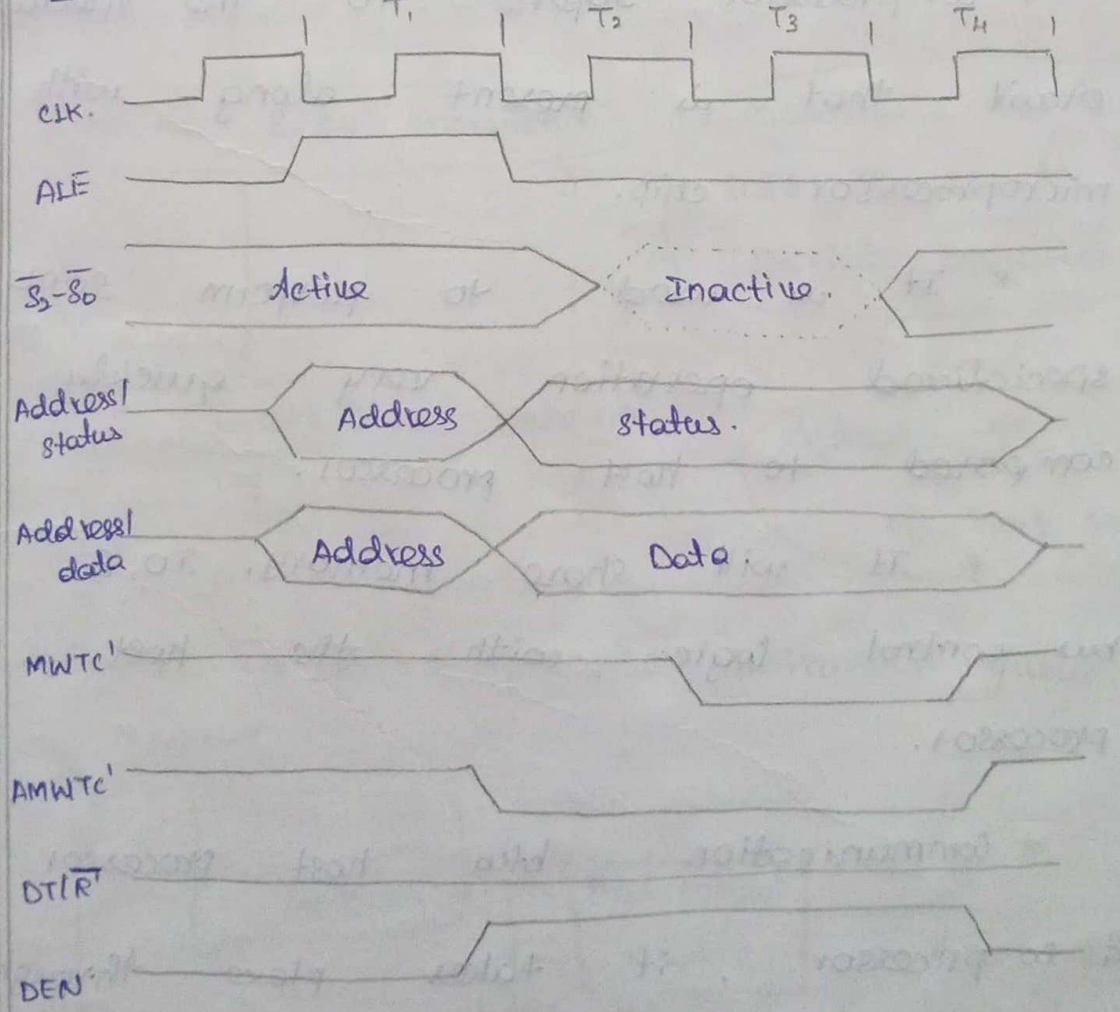
Disadvantages:

1. complexity is high. (or) $\uparrow$ se.

Read cycle:



write cycle:



MULTI PROCESSOR:

⇒ Multiple set of processor, executes instructions simultaneously.

⇒ It is possible only in MAX mode.

Multiprocessor configuration:

1. Co-processor.
2. closely coupled configuration.
3. loosely coupled configuration.

CO-processor Configuration:

* Co-processor refers to the specialized circuit that is present along with host microprocessor chip.

* It is used to perform some specialized operation very quickly compared to host processor.

* It will share memory, IO, & bus control logic with the host processor.

* Communication b/w host - processor & co-processor, it takes place through 4 signals, are

$\overline{\text{TEST}}$, RD/WT , Q_{s0} , Q_{s1}

⇒ $\overline{\text{TEST}}$ pin of host is connected to busy pin of co-processor.

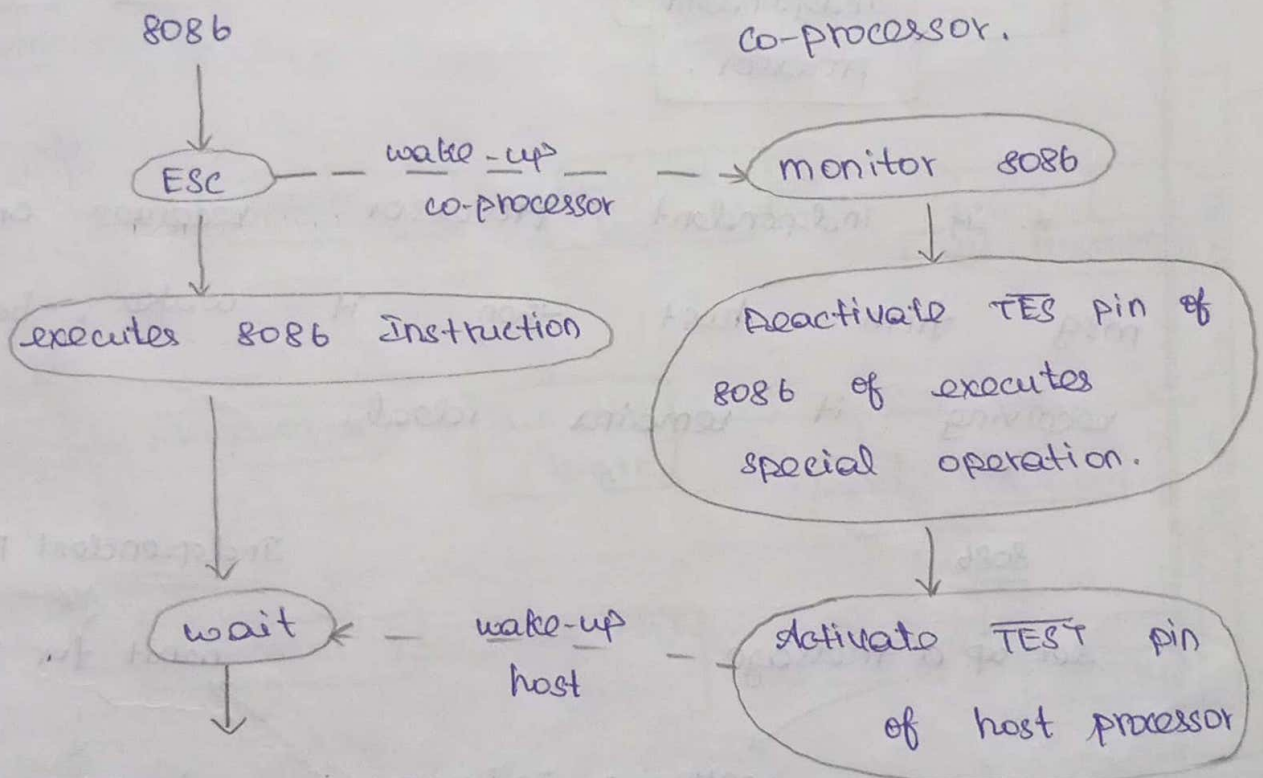
⇒ RD/WT , Q_{80} , Q_{81} of host connected to RD/WT , Q_{80} , Q_{81} of co-processor

* In co-processor we use two instruction that is.

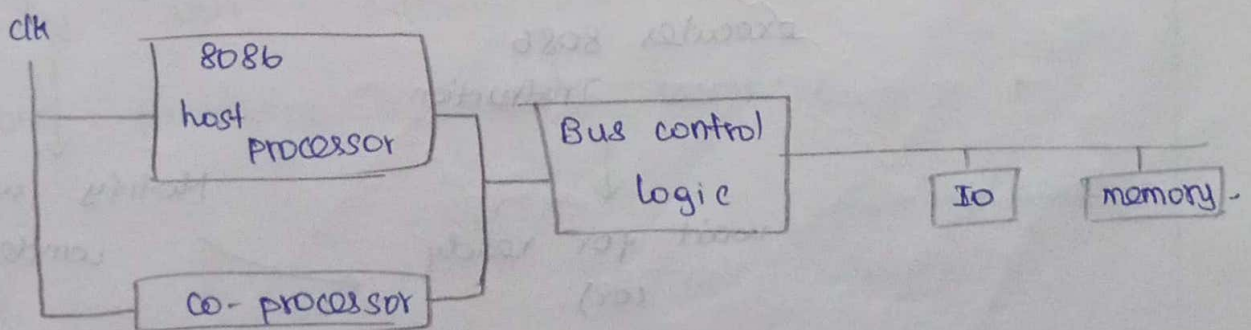
1. Escape instruction. (ESC)

2. wait instruction. (wait).

Flow Diagram:

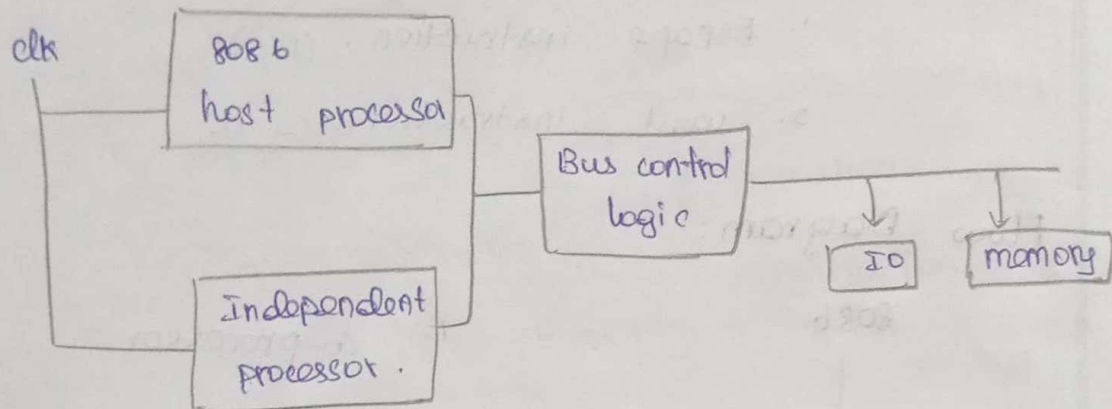


⇒ Q_{80} , Q_{81} → status of the instruction queue.

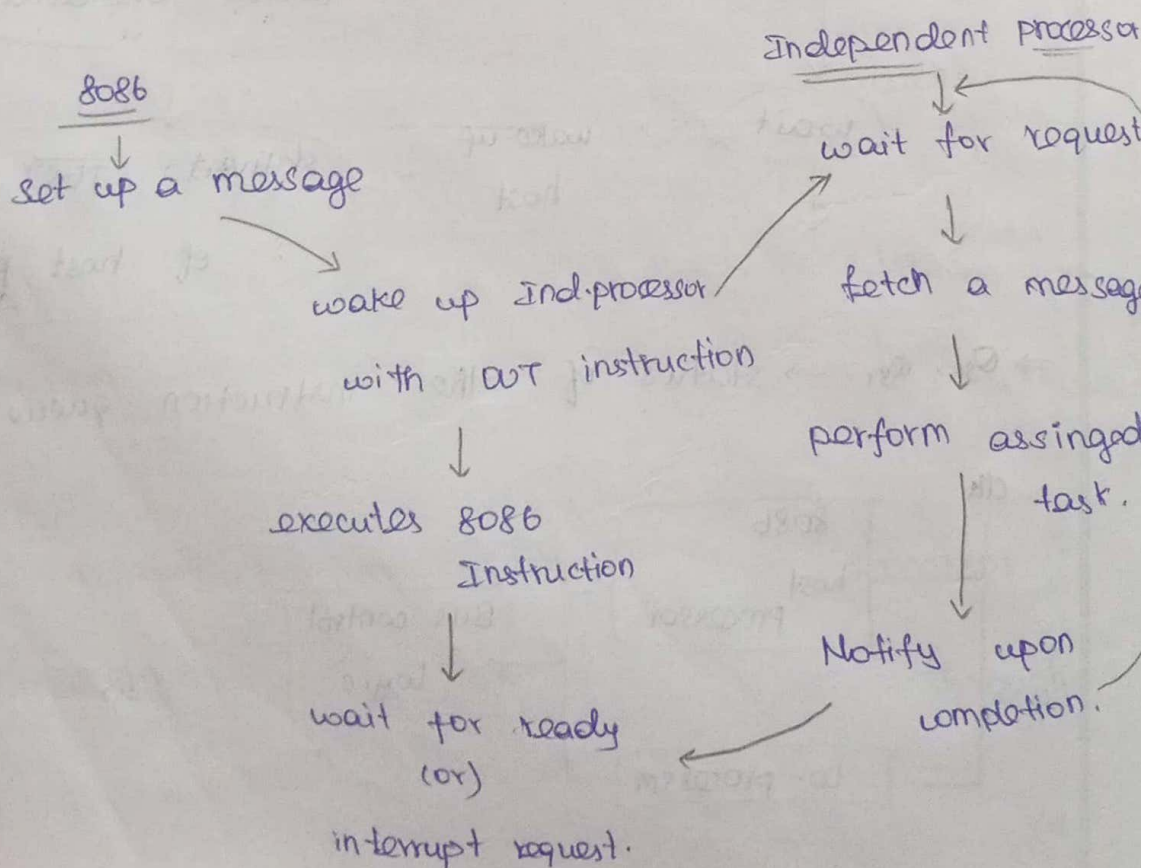


Closely - Coupled configuration.

*. The communication b/w host processor and independent processor, takes place through shared memory space.



*. If independent processor receives any msg from host then it wake, before receiving it remains ideal.



similarities:

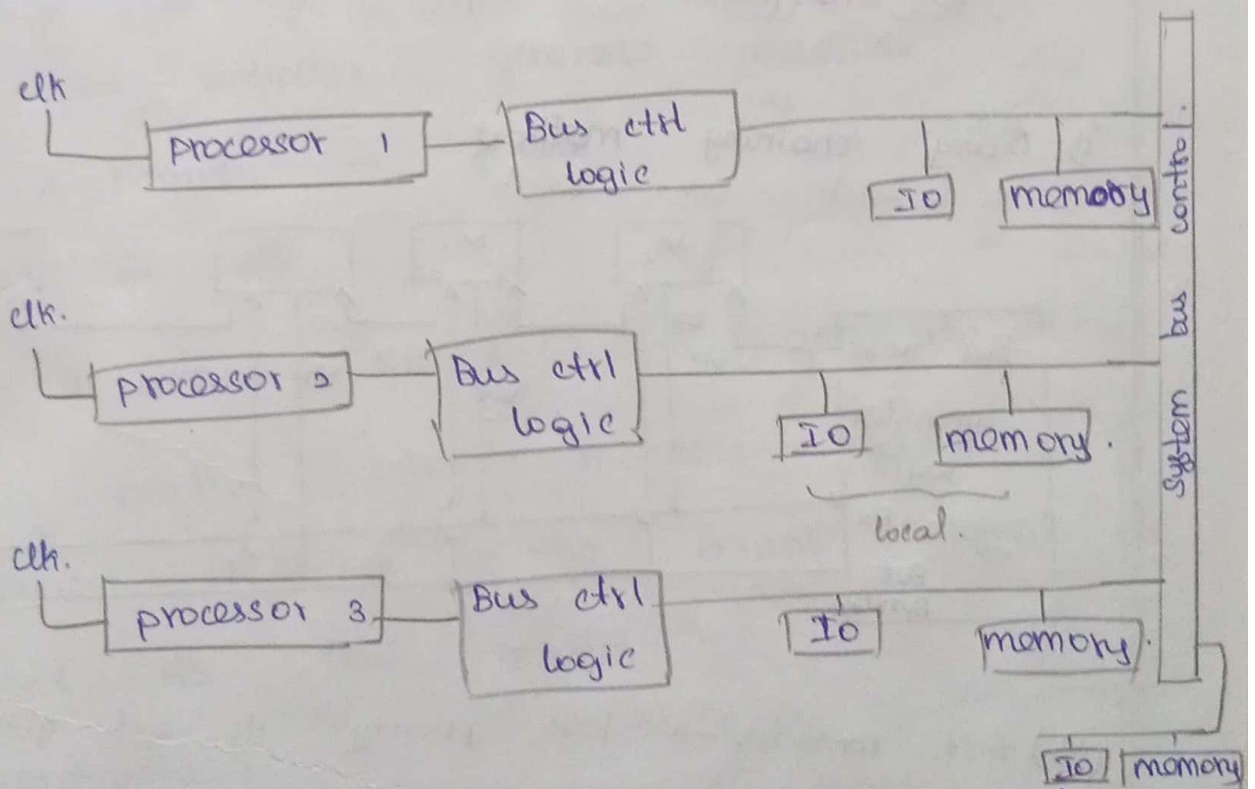
* here sharing io, memory, bus ^{control} ent
btw host & co-processor (or) ind. proc.

differences:

* In co-processor we use instruction
in closely-coupled we use memory space.

loosely-coupled configuration.

* Any no. of 8086 ^(or) other processor can be
connected.



* The resources can not be shared ^{global}
by any processor, it is specific to
one processor - Local resources.

Advantages:

* Any no. of modules can be added (or) removed without affecting the remaining system.

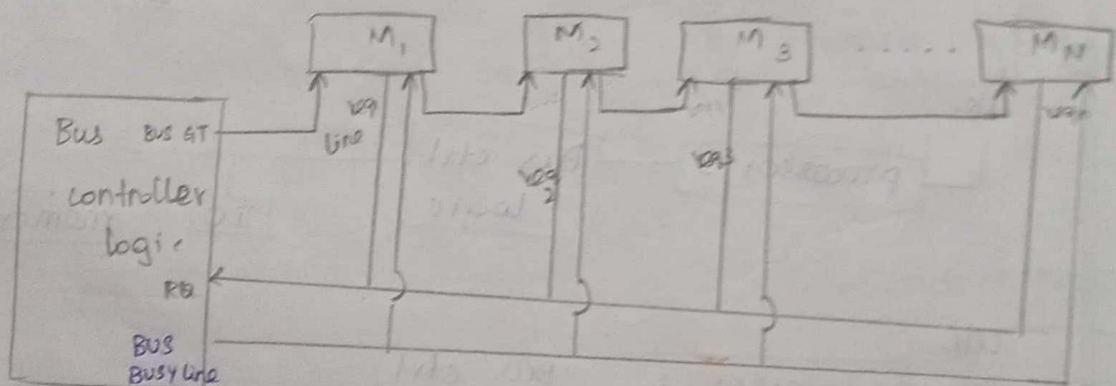
* High throughput

1). Daisy chaining method

2). Polling method

3). Independent request method

1). Daisy chaining method.



(M_1) first connected $\rightarrow$ high priority to get granted.

(M_N) last connected master $\rightarrow$ least priority

Advantage:

* Method is simply.

Disadvantages:

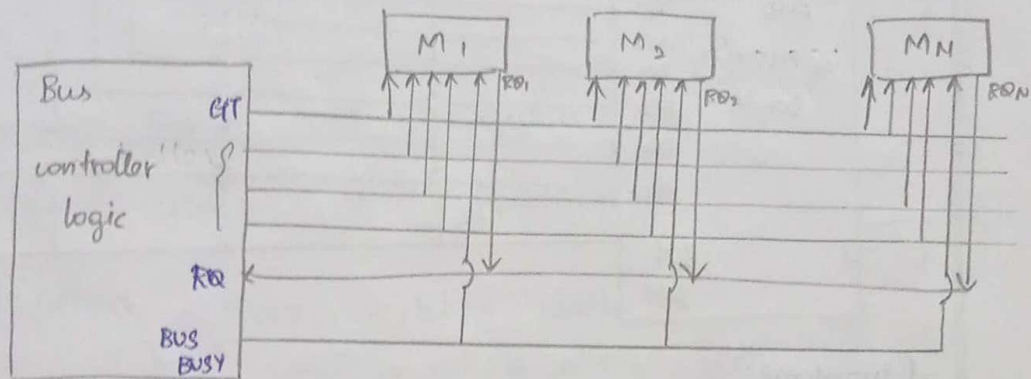
* Fixed priority.

* when system is large, propagation delay

is high,

→ Grant line is connected in chain manner.

a). Polling method:



* Bus controller generate address.

* Priority is fixed, 1st it check M_1 , last is M_N .

* Grant signal is generate by Bus control, when masters address & Request matches the grant signal is sent to that master.

Disadvantages:

* It requires more wires.

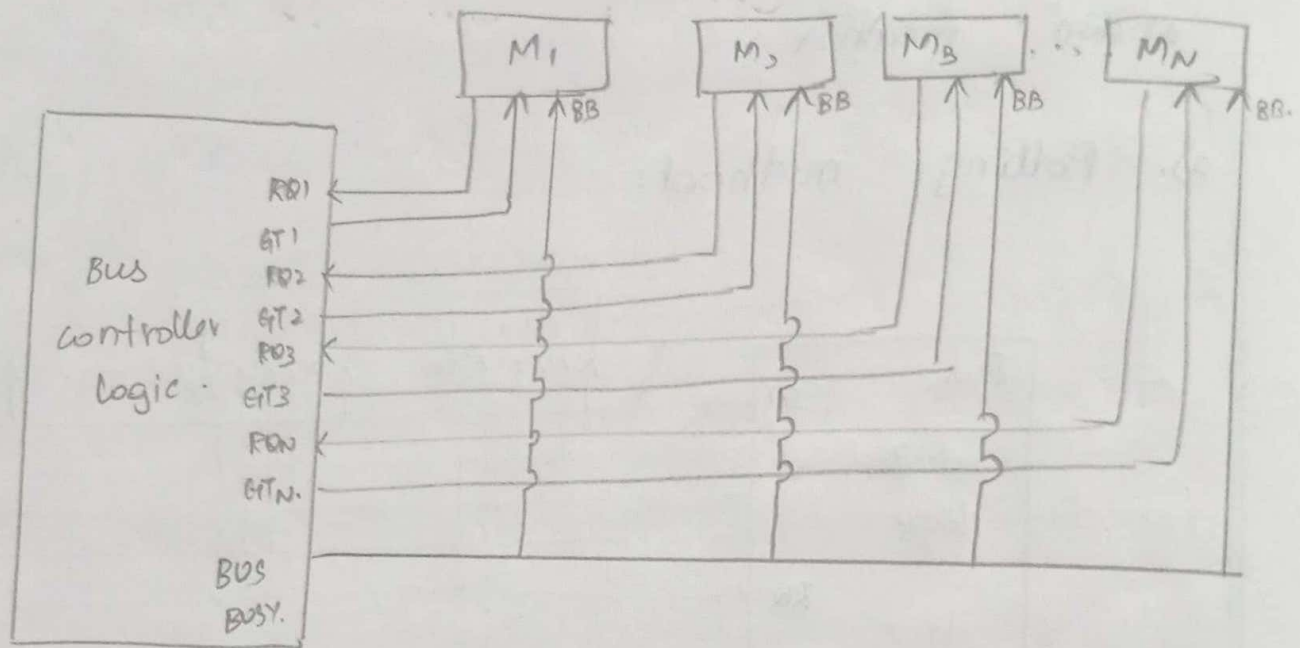
* Fixed priority.

Advantage:

* Individual master is problem

It does not affect (collapse) the entire system.

3) Independent Request Method:



Advantage:

- * Here use use rotating (or) random priority.
- * speed is high.